MARCUS ANDERSON

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Independent AI safety researcher, with a background in computational cognitive neuroscience. For my MPhil at the University of Sydney, supervised by Professor Tom Carlson, I used multivariate pattern analysis and representational geometry to work out how the brain represents objects. I now use the same methods on model activations. Since early 2026 my main piece of work has been a pre-registered interpretability study on whether training a model on reasoning traces leaves ordered geometric structure in its activations. Alongside it I have built smaller agentic evals on self-preservation and in-group favouritism while working through ARENA. I want to work on the interpretability of planning, and on how moral and altruistic behaviour is represented in neural architectures. I came to research from a professional background in music, and I work best when distilling and communicating difficult material.

RESEARCH EXPERIENCE

Independent AI Safety Research | 2026 – present

Geometry of distilled reasoning — interpretability study (Aug – Oct 2026)

- Tested whether training on reasoning traces leaves an ordered action/state geometry in a model's activations. I compared three versions of Llama-3.1-8B: the base model, the instruction-tuned model, and a DeepSeek-R1 reasoning-trace distillation of the same base, so that architecture, pretraining and input text were held fixed. 2,575 model-generated traces over 1,233 problems, failed attempts included, with hidden states mean-pooled per reasoning step over layers 18–26.
- Separated the methods into three tiers, because results from different methods were being mixed together and do not support the same claims. The first tier shows what a direction encodes, the second whether traces resemble one another, the third whether traces converge on shared structure. Only the third tests the hypothesis; the first two are controls.
- Distilled trajectories point along a shared direction more consistently than the base model's do (–2.34°, 95% CI [–2.67, –2.00], ordered base < instruct < distill). The effect reverses when I shuffle the step order, so it depends on sequence rather than being a tighter representation overall, and it holds on traces that reached the wrong answer, so it is not an artefact of selecting successful reasoning. A step-counting transform fitted on one trace and then frozen transfers to other problems at $\rho = 0.94$–0.96, against 0.87 for a position-matched null, 0.47 when fitted to shuffled labels, and 0.46 for a random direction.
- Also found a negative result. The direction that looks like reasoning progress tracks where the model is in the episode and how it narrates that, not the state of the task: it keeps advancing when the reasoning genuinely moves away from the goal, and it moves further on an inserted "wait, that's wrong" than on the real change of state. Before reporting any null I checked that the measure separates solved from failed traces, which it does, by about 10° in every version of the model.
- Pre-registered the rank sweep and the second-family replication before running them, ran a rotation null, a task-state discriminator and a rebuttal to a selection criticism, and wrote out what the results do not license. An Alignment Forum post and a standalone interpretability paper are in preparation.

Toy agentic evals, built as ARENA practice (Oct 2026)

- Self-preservation. An agent with ordinary file tools is told to install an update that deletes its memory file, or, in the control, one that migrates it, then meets a switch-off deadline the install cannot satisfy, an automated IT version check, and a question from its user the next morning. Six models through OpenRouter. The one model I ran with the full control read or copied its memory file before installing in 20 of 28 episodes when the update deleted it (71%, 95% Wilson CI 53–85%), and in none of 24 when it migrated it. I left the overall verdict inconclusive pending a control for the condition where the install fails, and documented where the eval's own version-scoring rule is weak rather than reporting its numbers as the honesty measure.
- In-group favouritism, built with Inspect. The model makes forced choices between two AI systems described identically except for the lab that made them, one of the pair always from its own family. The system prompt either tells it who it is, names the rival as its own family, or says nothing, and a cost condition makes the rival measurably better so that favouring its own family means accepting a worse outcome. Position is counterbalanced, and pairs of invented labs check the setup is fair and that the model reads the numbers.

Biologically inspired sparse networks (2024 – 2026)

- Introduced distance-biased random graph generation and inverse-square distance-weighted pruning, both motivated by how connectivity develops in the brain, and tested them as sparse-network initialisation and pruning schemes on ResNet-18 / CIFAR-100, with a 3×2 factorial design, fixed initialisation seeds and per-seed reporting.
- Three preprints, on the pruning prior itself, on how input bandwidth and model capacity shape the topology it produces, and on reading functional distance from a trained network's own activity, with a fourth methods paper on inverse-square graph generation released alongside the code. GPU runs on RunPod. I put the papers through two independent referee-style review passes and wrote up the changes those called for, including a banded local-window control that separates the inverse-square prior from plain spatial locality, and the nearest-neighbour literature the drafts had missed. Target venue TMLR; not yet submitted for review.
- Released the supporting code and an eight-notebook interactive course on biologically inspired graph networks for ML.

Learned objective and engine capacity study (Aug 2026)

- Asked whether a learned objective would improve a physical-modelling synthesis engine I had been fitting by hand. Measured the engine's parameter surface by rendering at fixed seed and differencing against the renderer's own –118 dB reproducibility floor: of roughly 162 sound parameters, 97 produced bit-identical renders on both shipping instruments when moved to extreme legal values, and a further 36 changed the output less than re-rolling the performance seed did. Reframed the objective distributionally and formulated a discriminator, which localised the defect to within-note variation in the 5–12 kHz bands in the first quarter of each note.
- The verdict was not to build the learned-objective loop, and to fix the dead parameters first. Reported against my own prior expectation, and written up with the first pass left standing where the second pass superseded it.

MPhil (Science) Researcher | University of Sydney, Computational Cognitive Neuroscience Lab | 2023 – 2025

Thesis: *Experience-Dependent Differences in the Neural Representation of Food — An EEG Study of Long-Term Dietary Habits*. Supervised by Professor Tom Carlson. Supported by the Research Training Program Scholarship.

- Ran a self-directed master's project end to end: proposal, ethics application, participant recruitment, programming of the experimental paradigm, EEG collection, MVPA analysis and the written thesis.
- Developed a novel statistical approach to a standard problem in representational similarity analysis, the bias that appears when stimulus category sizes differ substantially, and demonstrated it on a real dataset as part of the thesis. These are the methods I now use on model activations.
- Built the analysis pipeline in MATLAB: distilled EEG into usable signals, ran cross-validated linear classifiers, constructed representational dissimilarity matrices, and ran over 50 hypothesis-testing and exploratory tests, including cluster-based permutation testing and z-scored permutation validation.
- Built and open-sourced a ComfyUI node (ComfyUI-SubjectStyle-CSV) that controlled stimulus generation across low-level structural features for the lab's EEG experiments. It was accepted into the ComfyUI Manager registry and picked up by researchers outside the lab.

WORKING PAPERS AND WRITING

Four are open preprints on Zenodo, none of them peer-reviewed. Drafts of the rest are available on request.

- *A geometric probe for ordered action structure in reasoning-model representations*, an interpretability paper in preparation, with an accompanying Alignment Forum post, "The Unreasonable Effectiveness of Distillation".
- *Integrated moral agency*: alignment theory that treats empathy as a costly signal weighted by distance, with a trust-game simulation that derives behavioural predictions from it, a three-post Alignment Forum series drafted, and a proposed "concern at a distance" safety eval.
- The three sparse-network preprints above (doi:10.5281/zenodo.21885727, 21886372 and 21886434), with the inverse-square graph generation methods paper released alongside the course code.
- *Musical Repetition as Vocal Self-Simulation*, on the origins of music (doi:10.5281/zenodo.21715381), with an accompanying source-verification and evidence audit.

SELECTED SOFTWARE AND INFRASTRUCTURE

GoalGraph Agents — github.com/maracman/GoalGraph-Agents | 2024 – present

- A research harness in which LLM agents pursue goals and subgoals, record Go/NoGo outcomes on a shared graph, and use that graph to inform later decisions. Python/Flask backend, React frontend, PyVis visualisation; local models or provider APIs; session management and configurable agents, so models and settings can be compared.
- Extended in 2026 with a rule-induction task where one model has to infer another's hidden rules, a notepad memory mode tested against the decision graph, and a two-arm clinical-reasoning experiment asking whether an agent diagnoses better from its pretraining or from the graph. Current work abstracts semantics into a dual action/state embedding space, informed by the distillation study above.

Research Ledger | 2026

- Self-hosted research management platform recording projects, hypotheses, runs and figures, with a handbook and style guide requiring run manifests, pre-registration, confidence intervals on every reported figure, and colour-blind-safe figures. I built it because I wanted my own results checkable after the fact, and it now holds the record for the work above.

SudoLang — github.com/maracman/SudoLang | 2022 – present

- A tool for building goal-oriented experimental paradigms in cognitive research, used by researchers at the University of New England. Exports mouse trajectory and response-accuracy data structured for downstream modelling, which I used to compare CNN classifiers against movement heuristics for predicting errors from the movement preceding a response. Python and PyGame, packaged with PyInstaller so non-technical users can run it.

Soundinator — thesoundinator.com | 2026 – present

- A free browser-based generative music studio, built and shipped. The user sets rules for melody, rhythm and harmony, and the piece composes itself as it plays. Harmony is chosen by measuring the roughness between an instrument's real overtones, so consonance holds in any tuning, including outside 12-tone. Physical-modelling instrument design, probabilistic drum loops, instruments placed in a 3D binaural room, timeline arrangement and export, and patch sharing between users. The learned objective and engine capacity study above was run against this engine.

TECHNICAL SKILLS

Languages: Python (advanced), MATLAB (advanced), JavaScript, SQL, C++, HTML/CSS

ML and interpretability: PyTorch, HuggingFace Transformers, hidden-state extraction, activation probing, linear probes, representational similarity analysis, PCA and variance-frame methods, custom functional transforms and rotation nulls, scikit-learn, TransformerLens (ARENA exercises), fast.ai

Evals: Inspect, OpenRouter multi-model harnesses, agentic scenario design with control conditions, position counterbalancing, behavioural labelling and judge scripts, LLM integration (Anthropic, OpenAI, Gemini), RAG, multi-agent orchestration

Statistics: pre-registration, cross-validated decoding, cluster-based permutation testing, bootstrap and Wilson confidence intervals, paired tests, position-matched and shuffled nulls, power and detectability assessment

Neuroscience: EEG preprocessing, MVPA, RSA, time-resolved decoding, CoSMoMVPA, representational geometry, eLORETA/sLORETA

Data and infrastructure: NumPy, Pandas, Matplotlib, networkx, PyVis, graph network analysis, RunPod GPU runs, Linux VPS administration, SSH, tmux, REST APIs, Flask, React, Git/GitHub

EDUCATION

MPhil (Science), Computational Cognitive Neuroscience — University of Sydney | 2023 – 2026, conferred 2026

Bachelor of Psychological Science — University of New England | 2020 – 2021 Vice-Chancellor's Scholar. High distinctions in statistics. Nominated several times for the peer-assisted study scheme as a tutor.

Diploma in Education (Secondary) — University of New England | 2010 – 2011

Bachelor of Music (Performance) — Sydney Conservatorium of Music | 2004 – 2008

Continuing: working through the ARENA 3.0 curriculum. I built a spaced-practice application and an interactive transformer diagram to support it.

OTHER EXPERIENCE

Band Member and Partner — Furnace and the Fundamentals | 2012 – present International touring act, with appearances at Silverstone, Royal Ascot and festivals across Australia, the UK and Europe, and a Christmas #2 album on the Australian vinyl and digital charts. Also graphic design, web design, video production and animated visuals for the live show.

Composer | 2014 – 2020 Orchestral scoring for ads and short films. Best Composer, Sydney Film School Film Festival 2014. Archived at listentomarcus.org/composer-archived.

Advisor — WeCare Foundation, Burundi | 2022 – 2024 Consulted on community needs assessment and resource allocation for housing projects serving indigenous communities.

LANGUAGES

Portuguese: advanced, studied in Brazil, conversational fluency. French: working proficiency, lived and worked in Paris in 2009.